A MAJOR PROJECT REPORT ON

CAREERNEST: AI-POWERED COURSE RECOMMENDATION & MENTORSHIP PORTAL

A Centralized Career Intelligence Operating System for Engineering Students

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Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC SESSION 2025 – 2026

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE OF APPROVAL

This is to certify that the project entitled **"CareerNest: AI-Powered Course Recommendation & Mentorship Portal"** is a bona fide record of major project work carried out by **Akshat Kumar**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering** during the academic year 2025–2026.

The results embodied in this report have been verified through automated unit, integration, and security test suites, and have not been submitted to any other Institute or University for the award of any degree or diploma.

_____	_____	_____
[Project Supervisor / Guide]	**[Head of Department]**	**External Examiner**
Internal Guide	Head of Department (CSE)	Departmental Viva-Voce
Dept. of Computer Science & Engg.	Dept. of Computer Science & Engg.	Date: _____

DECLARATION

I hereby declare that the project work presented in this report entitled **"CareerNest"** is an authentic record of my own work conducted under the supervision of the Department of Computer Science & Engineering. All source code, architectural designs, and empirical benchmark results are original, and relevant literature citations have been duly acknowledged.

Akshat Kumar
Date: October 27, 2026

ABSTRACT

The contemporary software engineering hiring landscape presents an acute challenge for computer science undergraduates: while academic curricula emphasize discrete mathematics and foundational algorithmic paradigms, modern technical recruitment demands verifiable full-stack engineering proficiency, Applicant Tracking System (ATS) resume compatibility, behavioral fluency, and production deployment experience. Students are forced to context-switch across five or more disjoint platforms with zero unified feedback.

CareerNest is an integrated full-stack **Career Intelligence Operating System** engineered to eliminate this fragmentation. Built on the MERN stack (MongoDB, Express.js, React 18, Node.js) paired with Google Gemini 1.5 Flash Large Language Models, WebRTC video calling, and Socket.IO duplex websockets, CareerNest establishes a closed feedback loop: *Assess* → *Plan* → *Learn* → *Practice* → *Mentor* → *Apply* → *Verify*.

Key system innovations include: (1) an empirical **7-Factor Career Readiness Calibrator** computing a dynamic 0–100% preparedness index, (2) a **Dual-Engine AI Pipeline** with automatic heuristic failover guaranteeing 100% operational uptime, (3) an in-browser **Document Text Extraction & ATS Resume Scanner** (PDF/DOCX), (4) a **Verifiable Career Passport** governed by concurrency-safe partial unique compound database indices, and (5) a **Real-Time WebRTC Mentorship Suite**. The entire platform is verified by 124 automated tests (36 client + 88 server) and deployed live to production on Vercel with MongoDB Atlas cloud clustering.

PROJECT REPOSITORY & PRODUCTION LINKS

Artifact / Endpoint	URL / Repository Target	Status
GitHub Source Repository	https://github.com/Akshat-021/CareerNest.git (Branch: main)	**Public / Active**
Production Client (SPA)	https://career-nest-frontend-three.vercel.app	**Live on Vercel Edge**
Production API Gateway	https://career-nest-backend-three.vercel.app/api/health	**Healthy (DB Connected)**
Cloud Database Cluster	MongoDB Atlas AWS (cluster0.vcjhuih.mongodb.net)	**Replica Set Active**

CHAPTER 1: PROBLEM STATEMENT & REQUIREMENTS SPECIFICATION

1.1 The Employability Disconnect: While global tech hiring increasingly screens resumes via automated ATS algorithms and requires verifiable project proof-of-work, students lack actionable data on why their applications fail. Existing platforms solve isolated sub-problems (LeetCode for algorithms, Coursera for video lectures, Jobscan for ATS keywords), creating cognitive overload and disjointed skill profiles.

1.2 Software Requirements Specification (SRS):

- **FR-1 (Authentication & RBAC):** Email/password registration with bcrypt-10 encryption, 6-digit cryptographic OTP verification via Gmail SMTP, and role enforcement (Student, Mentor, Admin).
- **FR-2 (7-Factor Readiness Engine):** Atomic recalculation of multi-dimensional preparedness score on every user activity.
- **FR-3 (Document Text Extraction & ATS):** In-memory extraction of PDF/DOCX resumes and semantic analysis via Gemini 1.5 Flash.
- **FR-4 (WebRTC 1-on-1 Mentorship):** Duplex audio/video calling with cryptographic session authorization and room lifecycle management.
- **FR-5 (Course Center & Certifications):** Module progression tracking and tamper-resistant certificate generation with partial unique indexing.
- **NFR-1 (Serverless Optimization):** Database connection promise caching to eliminate AWS Lambda cold-start buffering timeouts.
- **NFR-2 (Accessibility & Security):** WCAG AA compliant glassmorphism design tokens, IDOR prevention, and production OTP redaction.

CHAPTER 2: MATHEMATICAL FORMULATION & ALGORITHMIC CORE

2.1 The 7-Factor Career Readiness Scoring Formula:
Readiness is evaluated through an empirical, weighted multi-dimensional synthesis implemented in server/services/careerProfileService.js:

$$\text{ReadinessScore} = 0.22(S_1) + 0.18(S_2) + 0.20(S_3) + 0.15(S_4) + 0.15(S_5) + 0.05(S_6) + 0.05(S_7)$$

Where:

- S_1 **(Technical Depth - 22%):** Verified competencies across Frontend, Backend, Database, and DevOps categories.
- S_2 **(Coding Proficiency - 18%):** Score performance across interactive algorithmic quizzes and challenges.
- S_3 **(Project Depth - 20%):** Public GitHub repositories, total stargazers, language diversity, and roadmap milestones.
- S_4 **(Resume ATS Quality - 15%):** Gemini 1.5 Flash evaluation of bullet point impact, quantifiable metrics, and keyword match.
- S_5 **(AI Mock Interview - 15%):** Rubric scoring across technical accuracy and behavioral STAR methodology.
- S_6 **(Certifications - 5%):** Earned course completion certificates and verified badges.
- S_7 **(Communication Clarity - 5%):** Speech pacing, articulation, and clarity metrics.

2.2 Dual-Engine AI Architecture (Gemini 1.5 Flash & Heuristic Fallback):

To ensure zero downtime, CareerNest implements a dual-engine AI pipeline: When GEMINI_API_KEY is present, requests are processed by Google Gemini 1.5 Flash with structured JSON schema outputs. If the external LLM times out (>15s) or if the key is absent, the system seamlessly transitions to a deterministic keyword-density heuristic engine without throwing runtime exceptions.

CHAPTER 3: DATABASE ARCHITECTURE & CONCURRENCY SAFEGUARDS

3.1 Partial Unique Compound Indexing for Tamper-Proof Certificates:
A critical vulnerability in legacy certificate systems is the duplicate key race condition: when students double-click 'Claim Certificate', concurrent requests execute simultaneously. A standard unique index on { student: 1, course: 1 } fails when historical certificates have course: null, permanently blocking students from receiving subsequent legacy credentials. CareerNest resolves this via a **Partial Unique Compound Index** in server/models/Certificate.js:

```
CertificateSchema.index( { student: 1, course: 1 }, { unique: true, partialFilterExpression: { course: { $type: 'objectId' } } } ); // Idempotent duplicate-catch handler in courseController.js catch (err) { if (err.code === 11000) { return res.status(200).json({ success: true, certificate: await Certificate.findOne(...) }); } throw err; }
```

CHAPTER 4: SOFTWARE TESTING, BENCHMARKS & SECURITY VERIFICATION

CareerNest is validated by an automated test harness covering client and server domains. All 124 tests execute with 100% pass rates in under 1 second total runtime:

Verification Suite	Module / Subsystem Tested	Asserted Constraints	Result
Client Suite (36 Tests)	Design Tokens, Navigation, AI Disclosures, Gates	WCAG AA contrast, role routing, safe retries	**36 / 36 PASS**
Server Security (16 Tests)	Community Hub & Moderation Subsystem	IDOR prevention, member roles, report actions	**16 / 16 PASS**
Server Hardening (19 Tests)	Phase 5 Seeding, Atomic Readiness, Courses	Production seed lockout, duplicate race guard	**19 / 19 PASS**
Mentorship Room (8 Tests)	WebRTC Session Authorization Subsystem	Student/Mentor match, HTTP 403 / 410 rejection	**8 / 8 PASS**
Course Enrollment (9 Tests)	Gate 2 Progress Tracking & Certificate Claim	100% completion check, spoofed module rejection	**9 / 9 PASS**
Release Safety (7 Tests)	Phase 7.1 Production Redaction & Gmail SMTP	OTP suppression, PII masking, App Passwords	**7 / 7 PASS**
Partial Unique Index (8 Tests)	Phase 7.2 Concurrency & Legacy Coexistence	E11000 duplicate handling, null-course support	**8 / 8 PASS**

CHAPTER 5: CONCLUSION & FUTURE ENGINEERING ROADMAP

5.1 Summary of Contributions: CareerNest provides a unified, production-tested Career Intelligence Operating System for engineering graduates. The project successfully implements multi-dimensional readiness scoring, dual-engine LLM intelligence, in-browser document parsing, real-time WebRTC communications, and serverless architectural resilience.

5.2 Future Roadmap:

1. **Automated Speech-to-Text (STT) Analytics:** Integrate OpenAI Whisper API for real-time transcription and filler-word detection during AI Mock Interviews.
2. **Blockchain Credential Anchoring:** Anchor issued Career Passport certificates onto Layer-2 blockchains (Polygon) for non-fungible proof-of-competence.
3. **University LMS Interoperability:** Implement LTI 1.3 protocols to synchronize student readiness data directly into Canvas and Blackboard.

REFERENCES

1. Google DeepMind, *"Gemini 1.5: Multimodal LLM Architecture and Context Processing,"* Google Research, 2024.
2. Fielding, R. T., *"Architectural Styles and the Design of Network-based Software Architectures,"* Ph.D. Dissertation, UC Irvine, 2000.
3. Mozilla Foundation, *"WebRTC 1.0: Real-Time Communication Between Browsers,"* W3C Recommendation, 2023.
4. MongoDB Inc., *"Partial Indexes and Compound Constraint Mechanics in Sharded Clusters,"* MongoDB Technical Docs, 2025.